CL-2005: Database Systems

Lab # 6: DDL & DML

Task 1:

Scenario:

Create a schema for an e-commerce system to manage `Customers`, `Products`, `Orders`, and `OrderDetails`. The system must enforce data accuracy and ensure proper relationships between tables.:

Task Steps:

1. **Create the Database:**

   Name the database `ECommerce`.

2. **Create Tables:**

   Design the following tables:

   1. **Customers Table:** Add details such as `id`, `name`, `email`, `phone`, `address`, and `signup_date`.
   2. **Products Table:** Extend to include `id`, `name`, `description`, `price`, `stock`, `category`, and `rating`.
   3. **Orders Table**: Add `id`, `customer_id`, `order_date`, `status`, and `total_amount`.
   4. **OrderDetails Table:** Include `order_id`, `product_id`, `quantity`, `price_per_unit`, and `total_price`.
   5. **Payments Table**: Record payment details like `payment_id`, `order_id`, `payment_date`, `amount`, and `payment_method`.

3. **Constraints:**

   Apply appropriate constraints:

   - `NOT NULL` to ensure mandatory fields.
   - `PRIMARY KEY` for unique identification.
   - `FOREIGN KEY` to establish relationships between tables.
   - `CHECK` to ensure valid data (e.g., positive salaries, valid credit hours, positive capacity, salary limits).
   - `DEFAULT` values where applicable (e.g., default city as 'Unknown', default `status` as "Active").

- Ensure `stock` in the `Products` table cannot be negative and `quantity` in the `OrderDetails` table is at least 1.

4. **Modify Tables:**

- Drop column `rating` from Products table
- Add a column `discount` to the `Products` table.
- Set the `discount` column to a default value of `0`.
- Modify the data type of `price` in the Products table to DECIMAL(5, 2).

Now insert at least 5 records in each table

5. **Perform the followings tasks using SQL queries:**

1. Choose one product and update its category to Accessories.
2. Update order status to "Delayed" if order is overdue by 7 days.
3. Show **lastest 3 orders** placed with their **status** and **total_amount.**
4. Retrieve the unique categories of Products.
5. Update the price of all the products in the 'Electronics' category to decrease by 10%.
6. List product names that **do not have 'a' as the 2nd letter** and are **out of stock**.
7. Create **ArchivedCustomers** table and transfer all customers from `Customers` table who signed up **before 2023**.
8. Delete all customers from `Customers` table who were signed up before 2023.
9. For each product in the `Products` table, label products as
   - **`'Out of Stock'`** if `stock` is zero
   - **`'Low Stock'`** if `stock` is between 1 and 10
   - **`'Medium Stock'`** if `stock` is between 11 and 50
   - **`'High Stock'`** if `stock > 50`
10. Retrieve all customers from the `Customers` table who have placed at least one order (Use exists).
11. For each customer in the Customers table, show status of customer as `customer_status` that shows:
    - **`'Active Buyer'`** if the customer has at least one order in the Orders table
    - **`'Inactive Buyer'`** if the customer has never placed an order

6. **Truncate and Drop:**

- Truncate the `Customers` and `Orders` table.
- Drop the `OrderDetails` table.